

CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

Research Paper

Prepared from the completed Customer Segmentation & Churn Pattern Analytics in European Banking

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Abstract

Customer churn is an important challenge for the banking sector because the loss of existing customers can affect long-term customer relationships and retention performance. This study analyses customer churn patterns in European banking using a dataset of **10,000 customers**. The analysis combines exploratory data analysis with **K-Means customer segmentation** to identify customer groups with different levels of churn risk.

The findings show that **2,037 customers, representing 20.37% of the dataset, have churned**. Noticeable differences were observed across geography, age and customer activity. Germany recorded the highest churn rate among the three countries at **32.44%**, while customers aged 46–55 recorded the highest age-group churn rate at **50.57%**. Inactive customers also showed a higher churn rate (**26.85%**) compared with active customers (**14.27%**).

For customer segmentation, seven behavioral and financial variables were standardized and used in K-Means clustering. Six customer segments were identified, with the highest churn observed among **Inactive High-Balance Customers (32.44%)**, **Senior At-Risk Customers (32.05%)**, and **Multi-Product At-Risk Customers (31.40%)**. In particular, Inactive High-Balance Customers accounted for **31.52% of total observed churn**, making them the most important segment from a retention perspective.

The study concludes that a segmentation-based approach can help banks identify priority customer groups and design more focused retention strategies. The results can support targeted engagement, proactive customer support and data-driven retention planning. **The findings represent associations within the available dataset and should not be interpreted as evidence of direct causation.**

Introduction

Customer retention has become an important area of concern for the banking sector, as customers differ considerably in their demographic characteristics, financial profiles, product usage and level of engagement. Understanding which groups are more likely to leave can help banks identify retention priorities and develop more focused customer-engagement strategies. In this context, analyzing churn patterns together with customer segmentation provides a useful way of understanding how different customer profiles are associated with customer attrition. The project brief also focuses on developing a segmentation-driven understanding of churn across geography, demographics and financial profiles.

This study examines customer churn using a dataset of **10,000 European banking customers**. The dataset contains information on customers' credit scores, geography, gender, age, tenure, account balance, number of products, credit-card ownership, active membership status and estimated salary, along with the churn outcome represented by the Exited variable. The analysis first explores the distribution of churn and examines differences across geographic, demographic and financial characteristics. This provides a broader understanding of the customer groups in which churn is more frequently observed.

The analysis recorded **2,037 churned customers, representing an overall churn rate of 20.37%**. Noticeable differences were observed across several customer characteristics. Germany showed a higher observed churn rate than France and Spain, while older customer groups and inactive customers also showed comparatively higher churn. In contrast, variables such as credit-card ownership, tenure, estimated salary and credit score showed relatively limited differences in churn within this dataset.

To obtain a more detailed view of customer profiles, the study applies **K-Means clustering** using seven financial, demographic and behavioral variables: credit score, age, tenure, balance, number of products, estimated salary and active membership. After standardizing these variables, six customer segments were

developed. The resulting segments were examined according to their customer characteristics, size and observed churn rates. This segmentation highlights groups such as **Inactive High-Balance Customers, Senior At-Risk Customers and Multi-Product At-Risk Customers** as important areas for retention attention.

The purpose of this study is therefore not only to measure the overall level of churn, but also to understand **where churn is concentrated and which customer segments contribute most to it**. The findings are intended to support targeted retention strategies rather than a single approach for the entire customer base. The analysis is complemented by an interactive Streamlit dashboard that allows stakeholders to explore customer and segment-level churn patterns. Overall, the study provides a data-driven foundation for understanding customer churn and supporting more focused retention planning in European banking.

Dataset & Methodology

1. Dataset Description

The study uses a European banking customer dataset containing **10,000 customer records**. The dataset provides information on customers' demographic, financial and banking-related characteristics, along with an outcome variable indicating whether the customer exited the bank.

The main variables used in the analysis include **Credit Score, Geography, Gender, Age, Tenure, Balance, NumOfProducts, Has Cr Card, Is Active Member, Estimated Salary, and Exited**. Customer Id was treated as an identification field rather than an analytical variable. The Year variable was also not used for analysis because it remained constant in the dataset.

The target variable, **Exited**, represents customer churn, where a value of 1 indicates that the customer has exited and 0 indicates that the customer has remained with the bank. In the dataset, **2,037 customers were classified as churned**, giving an overall churn rate of **20.37%**.

The dataset supports the project's objective of examining churn across **geographical, demographic and financial profiles** and developing a segmentation-driven understanding of customer retention.

The major variables used in the study are:

Variable	Description
Customer Id	Unique customer identifier
Credit Score	Customer credit score
Geography	Customer's country
Gender	Customer gender
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
NumOfProducts	Number of banking products held
Has Cr Card	Credit-card ownership

Variable	Description
Is Active Member	Customer activity status
Estimated Salary	Estimated customer salary
Exited	Customer churn indicator

2. Data Preparation and Preprocessing

The dataset was first examined to understand its structure, variables and overall quality. A missing-value check was performed, and **no missing values were identified** in the available variables. The data was then prepared for exploratory analysis and customer segmentation.

For the exploratory analysis, additional categorical groups were created for selected numerical variables. **Age Group** was created to compare churn across different age ranges, while **Balance Group** and **Credit Score Group** were created to examine whether churn patterns varied across different financial ranges.

For segmentation, customer-identification fields were excluded because they do not provide meaningful information about customer characteristics. The segmentation analysis focused on seven variables:

- Credit Score
- Age
- Tenure
- Balance

- NumOfProducts
- Estimated Salary
- Is Active Member

These variables represent a combination of financial, demographic and behavioral characteristics.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify patterns in customer churn before performing segmentation. Churn rates were examined across **geography, gender, age group, number of products, active membership, credit-card ownership, tenure, balance group and credit-score group**.

The analysis also examined the distribution of estimated salary by churn status and calculated correlations between numerical variables and the churn outcome. This stage helped identify characteristics associated with higher or lower observed churn and provided the basis for interpreting the customer segments developed later.

4. Customer Segmentation Methodology

To identify groups of customers with similar characteristics, the study applied the **K-Means clustering algorithm**. K-Means was selected because the objective was to divide customers into groups based on similarities in their financial, demographic and behavioral characteristics.

Before clustering, the seven selected variables were standardized using **Standard Scaler**. Standardization was necessary because the variables were measured on different scales. For example, balance and salary have substantially larger numerical values than variables such as tenure or the number of products.

The scaled dataset was then evaluated using different numbers of clusters. Values of **K from 2 to 8** were tested using two measures:

- **Within-cluster inertia**
- **Silhouette score**

The highest silhouette score obtained was **0.165 for K = 6**. Therefore, six clusters were selected for the final segmentation analysis. The relatively low silhouette score indicates that the customer groups are not strongly separated; consequently, the clusters are interpreted as useful analytical profiles rather than completely distinct customer populations.

5. Segment Profiling and Churn Analysis

After assigning each customer to one of the six clusters, each segment was profiled using customer count, average age, balance, number of products, activity level and churn rate.

The six resulting customer segments were interpreted as:

1. **Active High-Balance Customers**
2. **Inactive Low-Balance Customers**
3. **Senior At-Risk Customers**
4. **Inactive High-Balance Customers**

5. **Active Low-Balance Loyal Customers**

6. **Multi-Product At-Risk Customers**

The churn rate of each segment was then calculated and compared with the overall churn rate. In addition to comparing **segment-level churn rates**, the analysis calculated each segment's **contribution to total observed churn**. This distinction was important because a segment can have a high churn rate but represent a relatively small share of the total customer base.

The methodology therefore combines **descriptive analysis and unsupervised customer segmentation** to identify where churn is concentrated and which customer groups should receive greater retention attention.

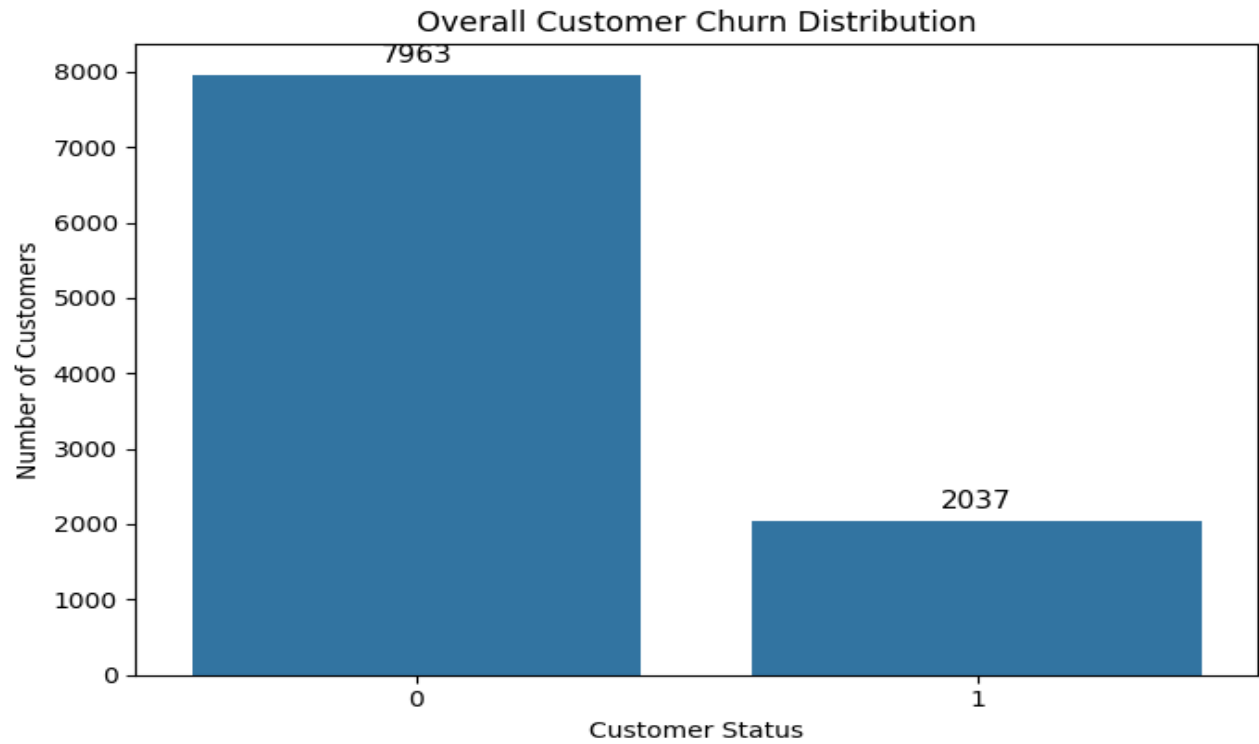
Exploratory Data Analysis

The Exploratory Data Analysis was carried out to understand the distribution of customer churn and identify patterns across demographic, geographic, financial and behavioral characteristics. The analysis was performed before customer segmentation so that the major characteristics associated with observed churn could be understood first. The project brief also identifies EDA and insights as key components of the research paper.

1. Overall Customer Churn

The dataset contains **10,000 customers**, of whom **2,037 customers have churned**, while **7,963 customers have remained with the bank**. This results in an overall churn rate of **20.37%**.

The result indicates that approximately one-fifth of the customers in the dataset have exited the bank. This provides the starting point for examining whether churn is concentrated within particular customer groups.



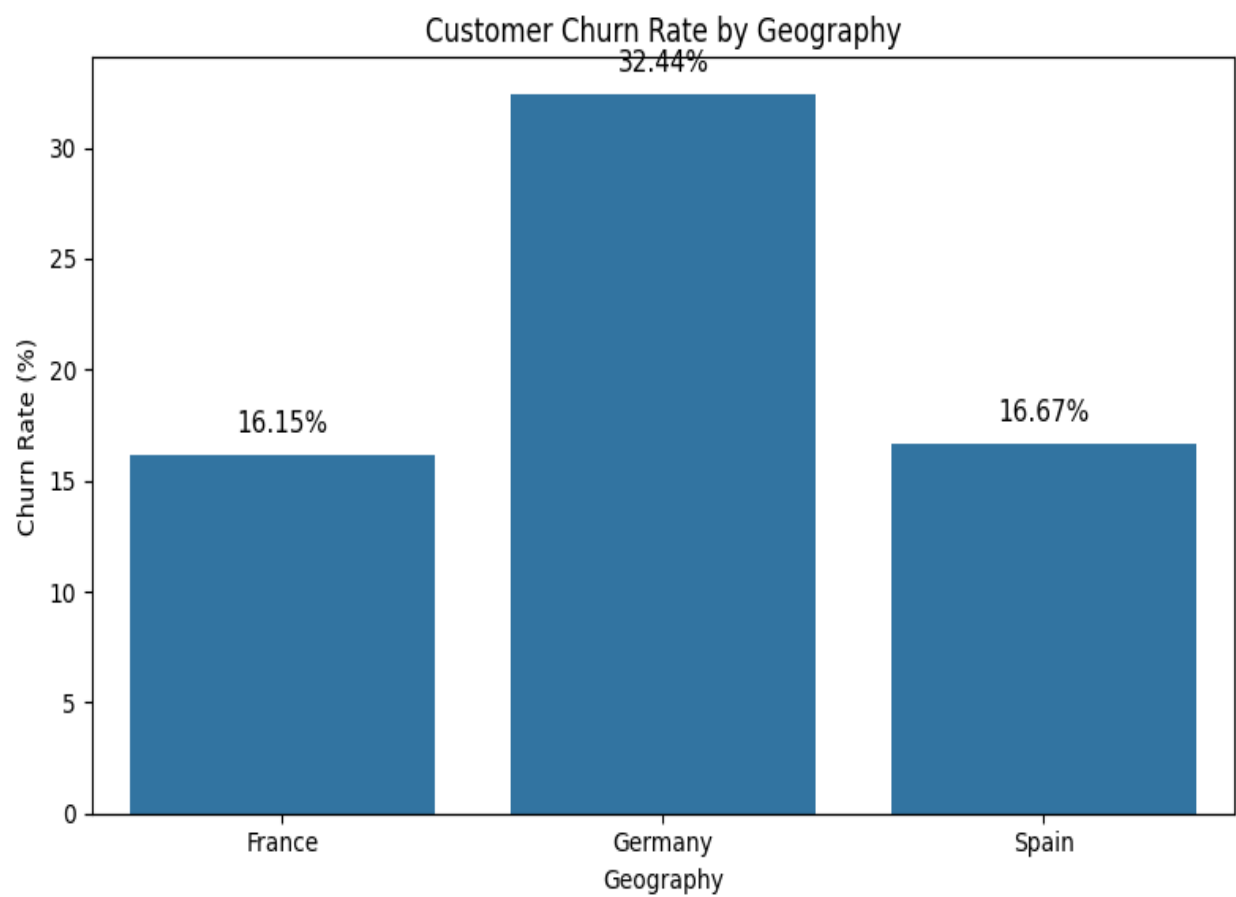
2. Churn Rate by Geography

Churn rates were compared across the three geographical markets represented in the dataset.

Geography	Churn Rate
France	16.15%
Germany	32.44%
Spain	16.67%

Germany recorded the highest churn rate at **32.44%**, whereas France and Spain recorded substantially lower rates of 16.15% and 16.67%, respectively.

The difference suggests that customer churn varies considerably across geographical markets. Therefore, retention strategies may need to consider market-specific customer characteristics and engagement patterns.

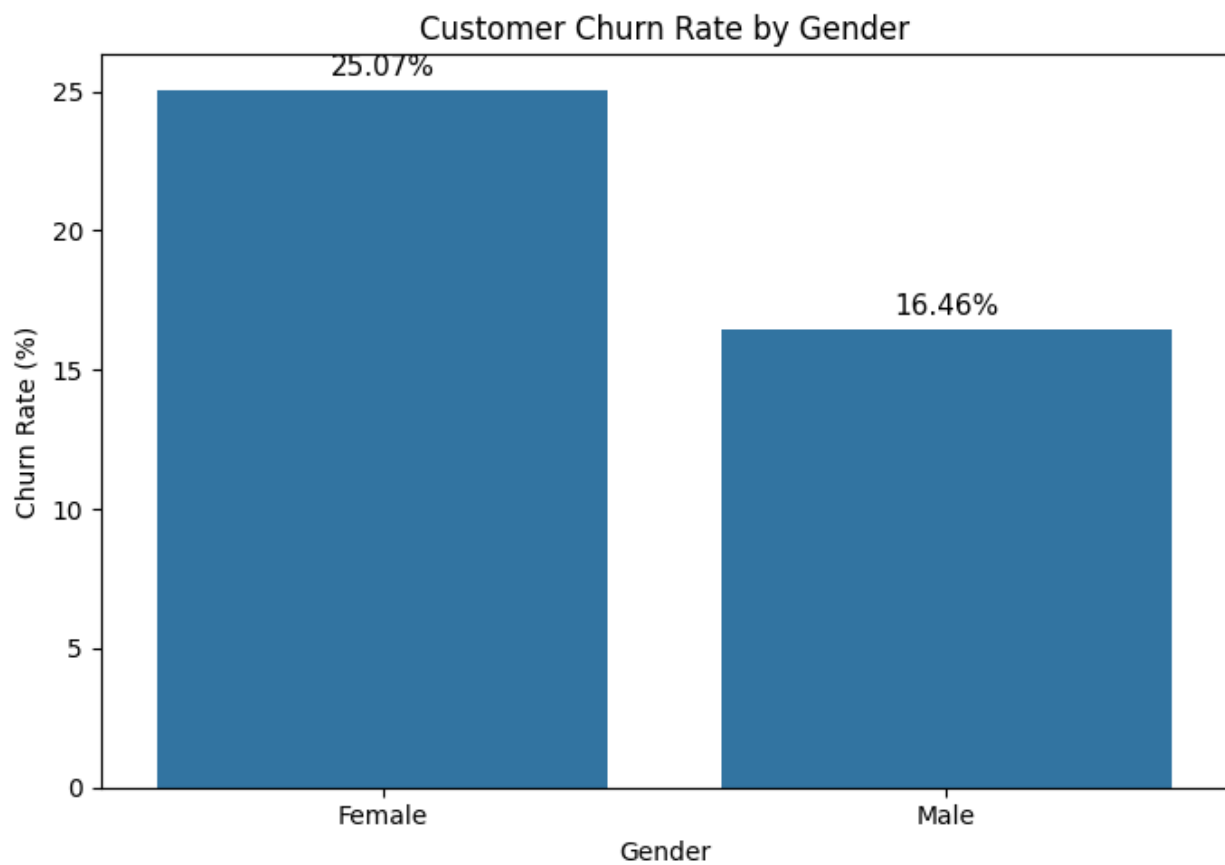


3. Churn Rate by Gender

The analysis also examined differences in churn between male and female customers.

Gender	Churn Rate
Female	25.07%
Male	16.46%

Female customers recorded a higher observed churn rate than male customers. The difference indicates variation in churn across the two groups within the dataset. However, gender should be treated primarily as a descriptive characteristic and should not by itself be used to determine individual customer treatment.



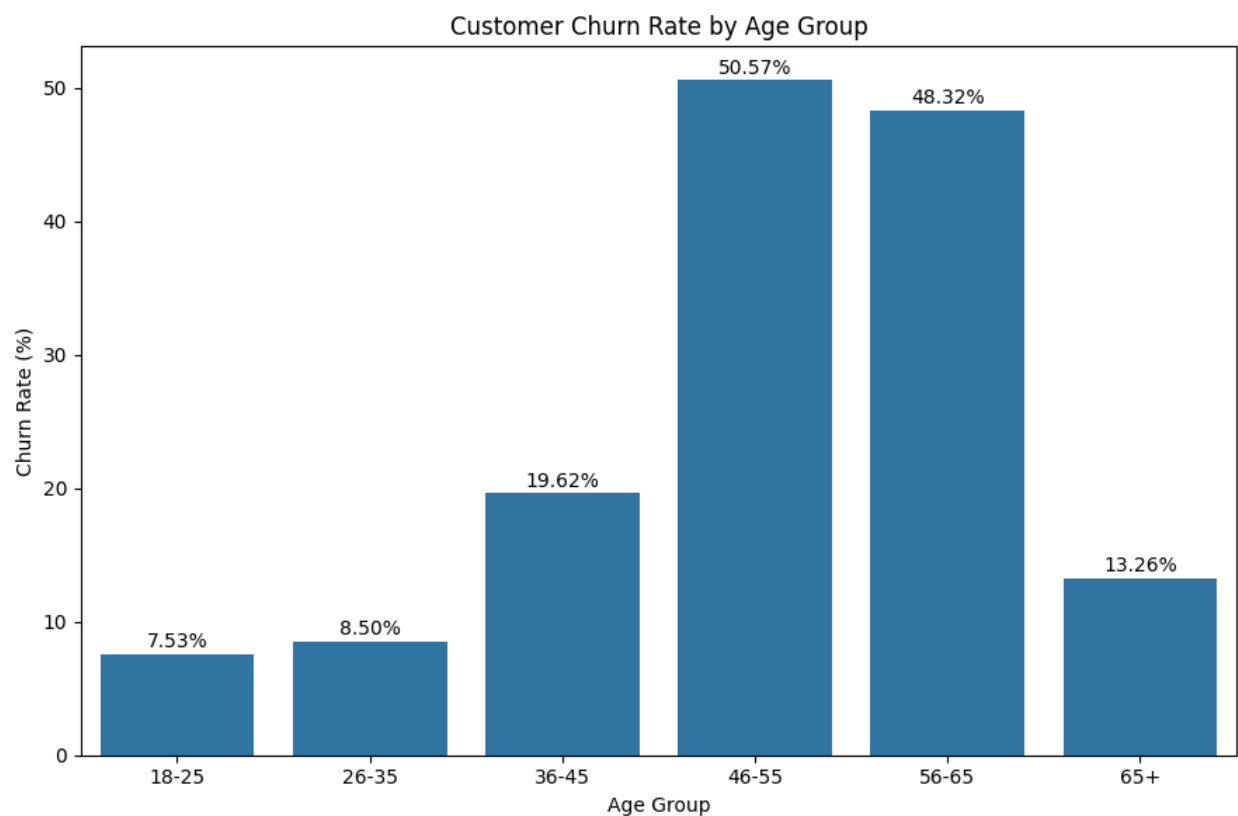
4. Churn Rate by Age Group

Age was divided into six groups to examine whether churn varied across different stages of the customer age distribution.

Age Group	Churn Rate
18–25	7.53%
26–35	8.50%
36–45	19.62%
46–55	50.57%
56–65	48.32%
66+	13.26%

A substantial difference can be observed across age groups. The **46–55 group recorded the highest churn rate at 50.57%**, followed by the 56–65 group at 48.32%. In comparison, customers aged 18–25 and 26–35 recorded much lower churn rates.

This pattern makes age an important characteristic for further investigation and is also reflected in the later customer segmentation analysis.



5. Churn Rate by Number of Products

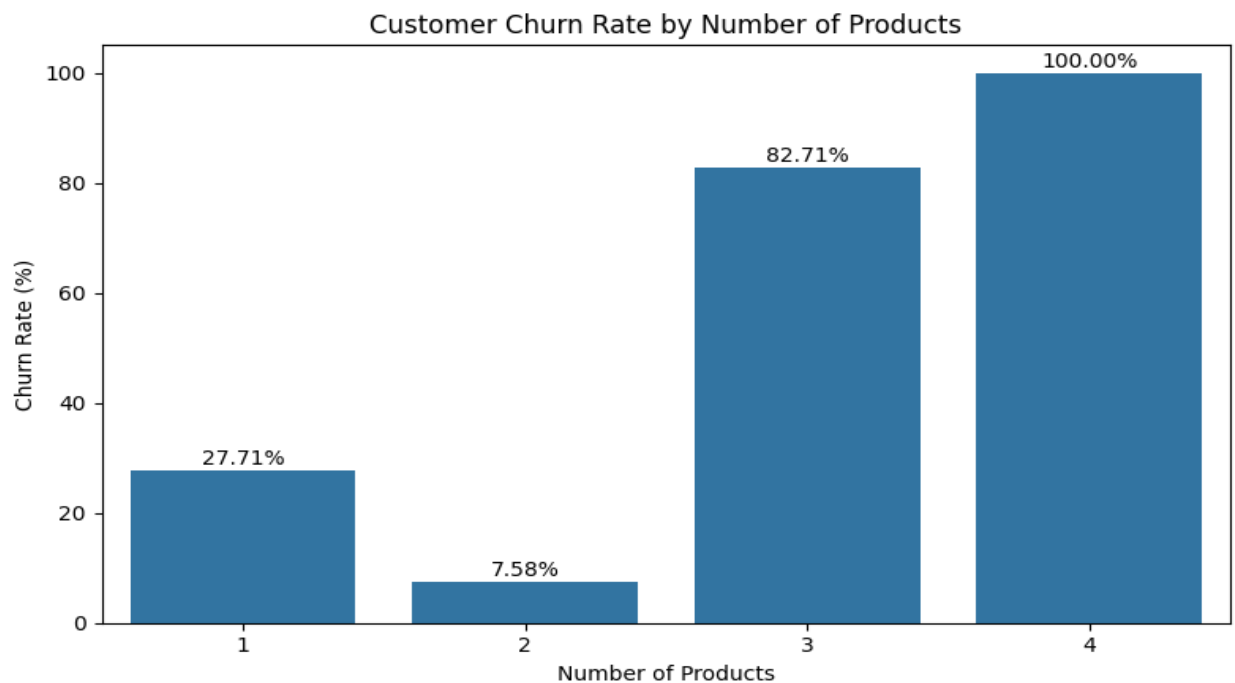
The relationship between the number of banking products and churn was examined.

Number of Products	Churn Rate
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1	27.71%
2	7.58%
3	82.71%
4	100.00%

Customers with three or four products show extremely high observed churn rates. However, these categories may contain relatively few observations. Consequently, these percentages should be interpreted cautiously and should not be treated as evidence that having more products directly increases churn.

The result nevertheless indicates that product ownership and utilization deserve consideration when analyzing customer profiles.



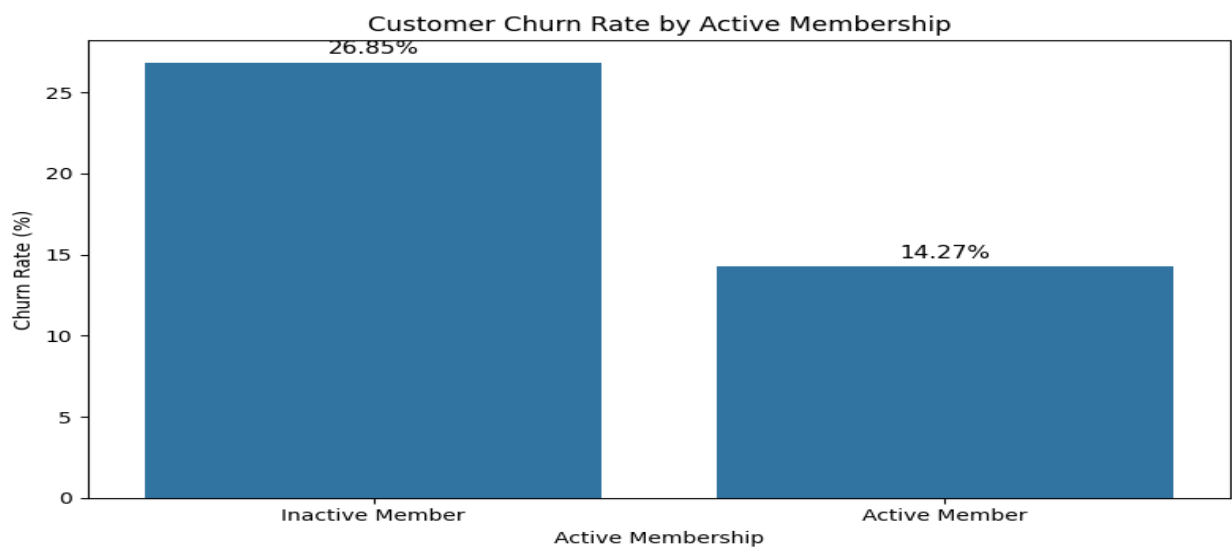
6. Churn Rate by Active Membership

Customer activity showed a clear difference in observed churn.

Membership Status	Churn Rate
Inactive	26.85%
Active	14.27%

Inactive customers recorded a substantially higher churn rate than active customers. The difference suggests that customer engagement is an important characteristic to consider when identifying groups requiring retention attention.

This observation also becomes important in the segmentation stage, where activity status contributes to the identification of high-risk customer profiles.

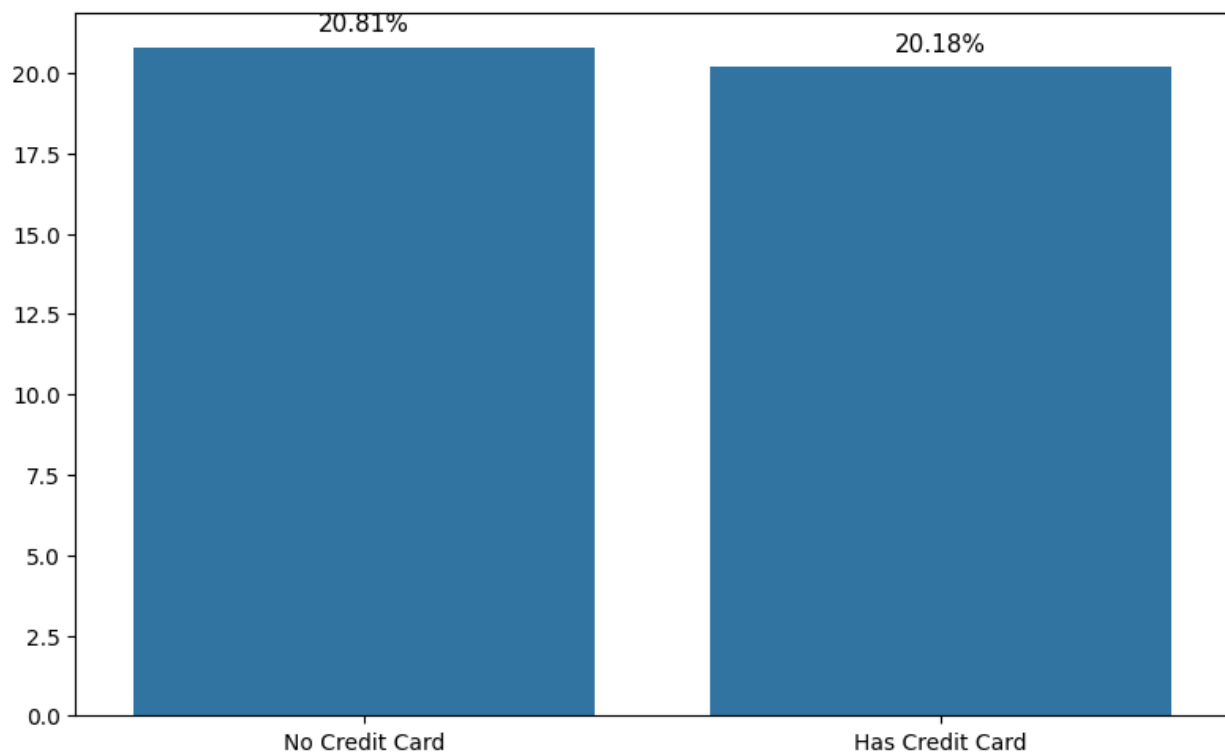


7. Churn Rate by Credit-Card Ownership

The analysis found very little difference between customers with and without a credit card.

Credit Card Ownership	Churn Rate
No	20.81%
Yes	20.18%

The difference is only 0.63 percentage points. Therefore, credit-card ownership does not appear to be a strong differentiator of churn in the analyzed dataset.

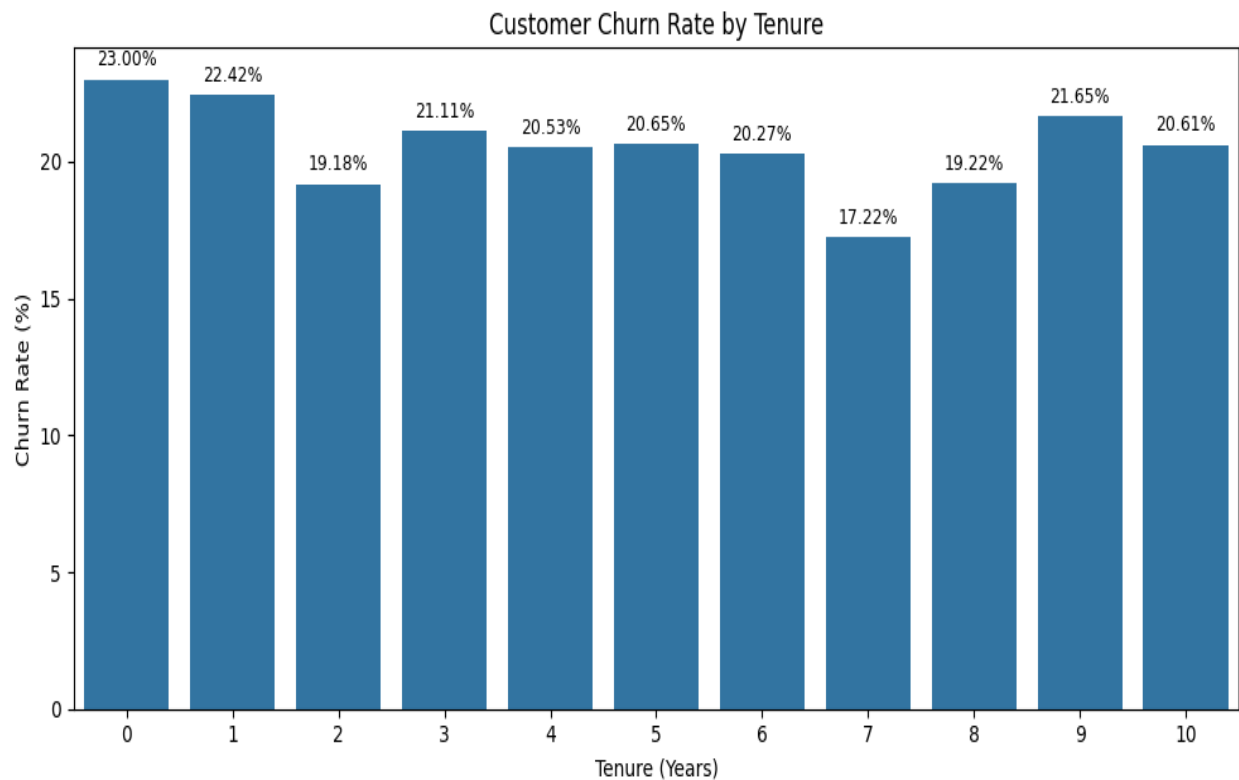


8. Churn Rate by Tenure

Churn rates were also examined across different customer-tenure levels.

The observed churn rates generally remained within approximately the **17%–23% range**, without a strong or consistent pattern.

This suggests that tenure alone does not provide a clear distinction between customers who churn and those who remain with the bank. The length of the customer relationship may therefore be more useful when considered together with other financial and behavioral characteristics.



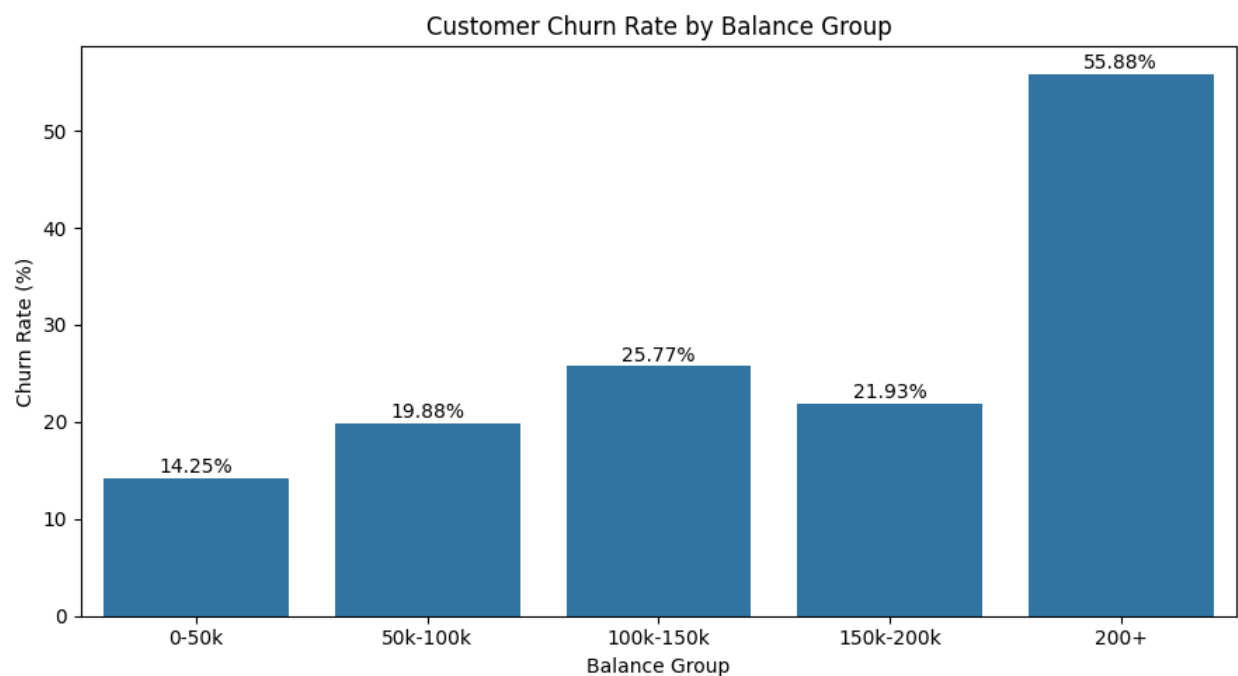
9. Churn Rate by Balance Group

Customer balances were divided into groups to examine whether financial position was associated with different levels of observed churn.

Balance Group	Churn Rate
0–50K	14.25%
50K–100K	19.88%
100K–150K	25.77%
150K–200K	21.93%
200K+	55.88%

The **200K+ balance group recorded the highest churn rate at 55.88%**. The 100K–150K group also showed a higher rate than the lower-balance categories.

However, the 200K+ category should be interpreted carefully because the number of customers in this group is relatively small. The finding is more meaningful when combined with customer activity and other characteristics, as demonstrated later through customer segmentation.

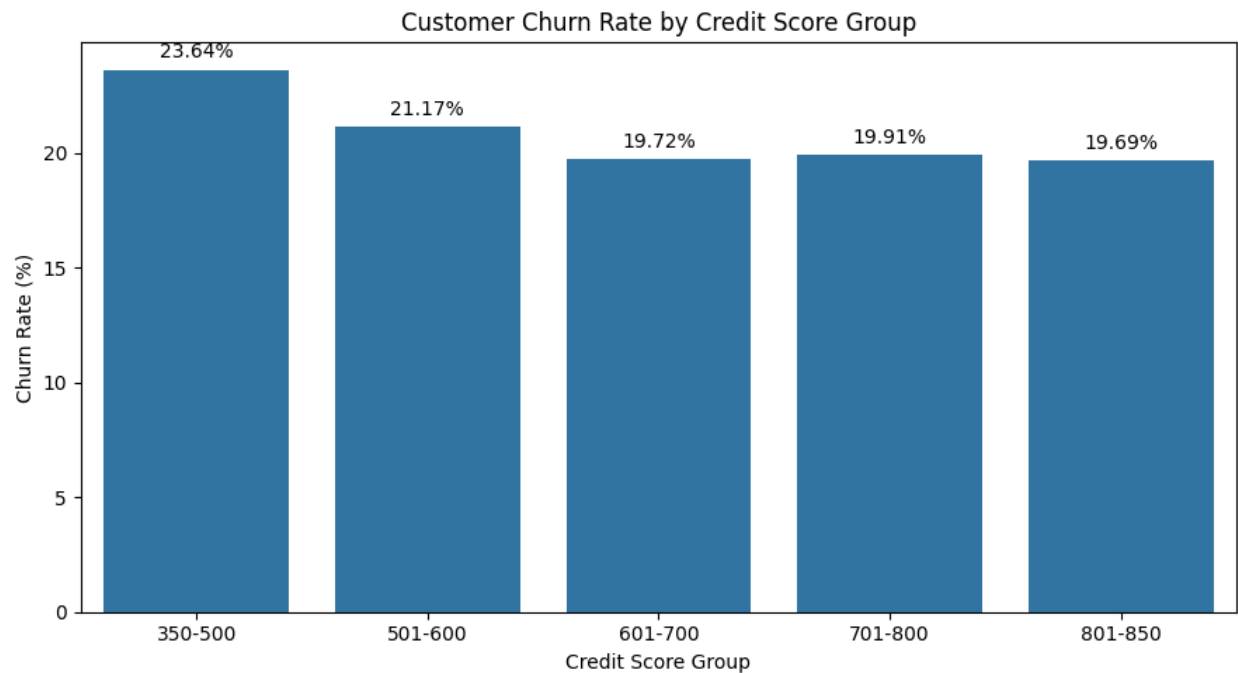


10. Churn Rate by Credit Score Group

Credit scores were grouped into five ranges.

Credit Score Group	Churn Rate
350–500	23.64%
501–600	21.17%
601–700	19.72%
701–800	19.91%
801–850	19.69%

The differences between most credit-score groups are relatively small. Although customers in the lowest credit-score group recorded somewhat higher churn, credit score does not appear to be a strong standalone discriminator of churn in this dataset.

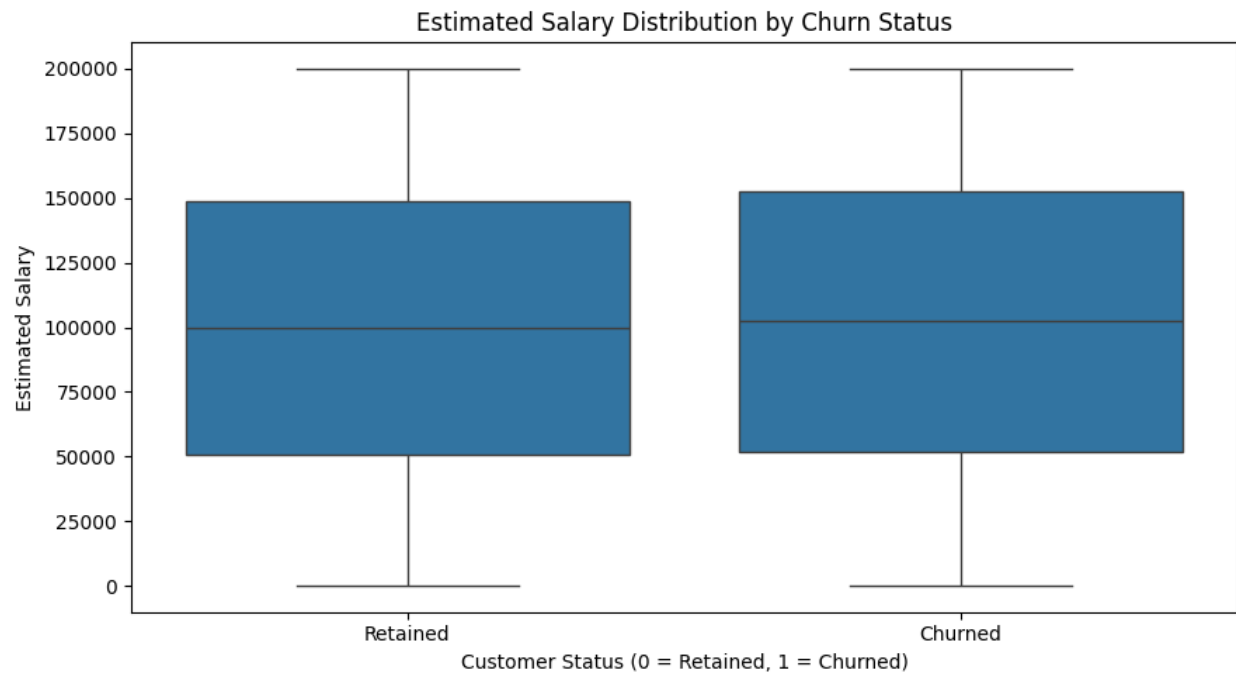


11. Estimated Salary and Churn

Estimated salary was compared between customers who churned and those who remained with the bank.

The distributions of estimated salary across the two churn-status groups were broadly similar. This indicates that estimated salary alone does not provide a strong separation between churned and retained customers.

Therefore, salary is more appropriately considered as one component of a wider customer profile rather than as an independent indicator of churn.



12. Correlation Analysis

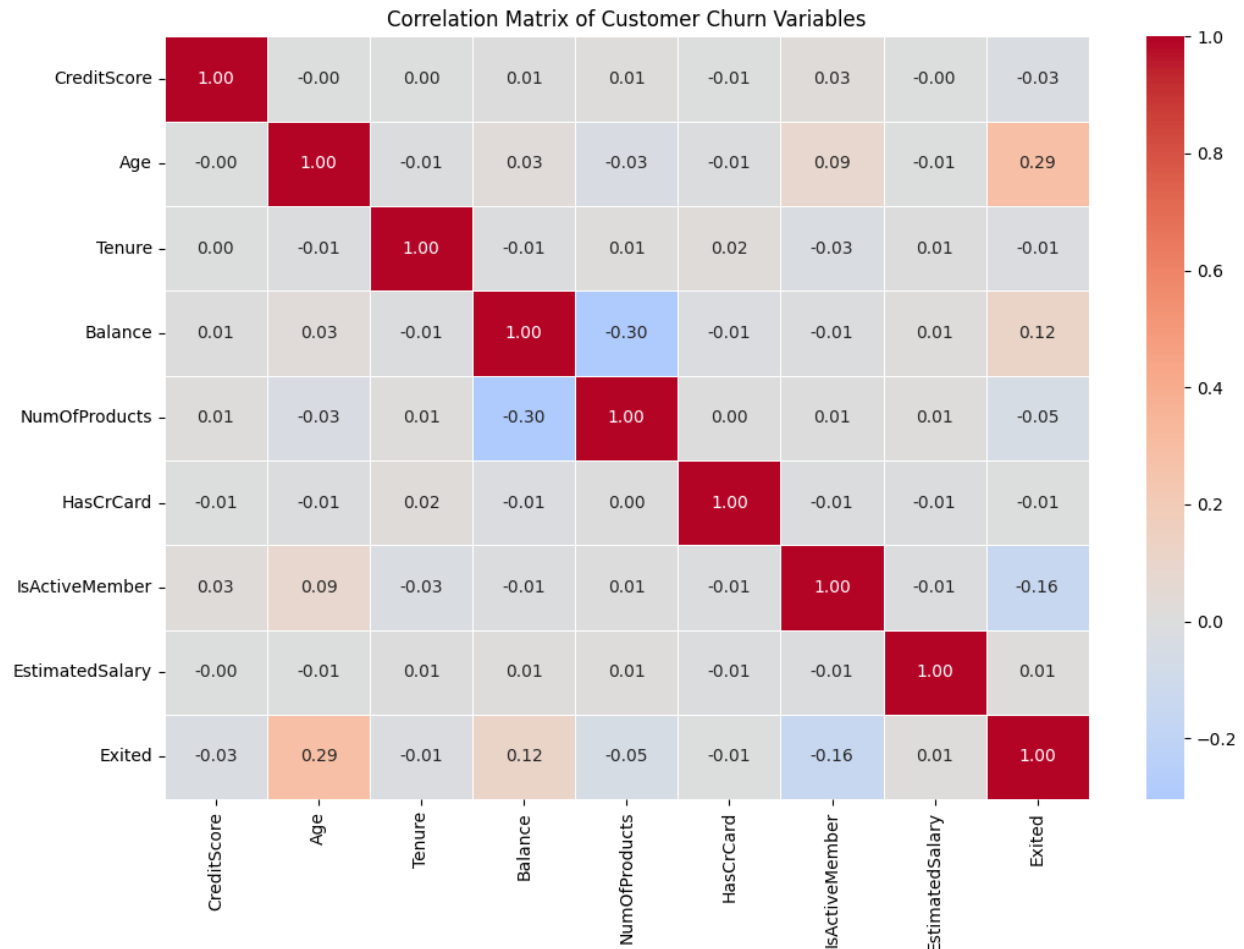
A correlation analysis was performed to examine the relationships between numerical customer characteristics and the churn outcome.

Variable	Correlation with Churn (Exited)
Age	+0.285
Balance	+0.119
Is Active Member	-0.156
Estimated Salary	+0.012

Variable	Correlation with Churn (Exited)
Tenure	-0.014
Credit Score	-0.027
Has Cr Card	-0.007
NumOfProducts	-0.048

Among the numerical variables, **Age showed the strongest positive correlation with churn (+0.285)**. Balance also showed a positive relationship, although considerably weaker. Active membership had a negative correlation of **-0.156**, which is consistent with the earlier finding that inactive customers have a higher observed churn rate.

The remaining variables showed relatively weak correlations with the churn outcome. Overall, the correlation analysis suggests that churn is unlikely to be adequately explained by a single numerical characteristic.



Overall EDA Findings

The EDA provides several important observations that guide the subsequent segmentation analysis.

- Overall customer churn is **20.37%**.
- **Germany** has the highest geographical churn rate at **32.44%**.
- The **46–55 age group** records the highest age-specific churn at **50.57%**.
- **Inactive customers** have a higher churn rate (**26.85%**) than active customers (**14.27%**).
- Very high churn percentages are observed among customers with three or four products, although these categories require cautious interpretation because of their potentially small sample sizes.

- Customers in the **200K+ balance group** show high observed churn, but this result also requires caution because of group size.
- Credit-card ownership, tenure, salary and credit score show comparatively weak differentiation in this dataset.
- The EDA therefore supports moving beyond individual variables and examining **combinations of customer characteristics through segmentation**.

Customer Segmentation

Customer segmentation was carried out to divide the bank's customers into groups with similar financial, demographic, and behavioral characteristics. The purpose was to move beyond overall churn rates and identify specific customer profiles that require different retention strategies. This approach supports the project objective of developing a segmentation-driven understanding of churn in European banking.

1. Segmentation Variables

Seven customer attributes were selected for segmentation:

- Credit Score
- Age
- Tenure
- Balance
- Number of Products

- Estimated Salary
- Active Membership

Customer Id was not used because it is an identifier rather than a customer characteristic, while Year was excluded because it remained constant across the dataset.

Before applying clustering, the selected variables were standardized using **Standard Scaler**. Standardization was necessary because the variables were measured on different scales. For example, balance and salary have much larger numerical values than tenure or the number of products.

2. K-Means Clustering

The **K-Means clustering algorithm** was used to identify groups of customers with similar characteristics.

Different values of K, ranging from 2 to 8, were evaluated using inertia and silhouette scores.

Number of Clusters (K)	Inertia	Silhouette Score
2	59,855.60	0.152
3	53,482.55	0.145
4	49,559.92	0.148
5	46,202.94	0.155
6	43,665.75	0.165
7	41,619.04	0.156
8	40,001.22	0.149

Based on the highest silhouette score, **six clusters were selected** for the final segmentation. The silhouette score of 0.165 indicates that the customer groups are not highly separated. Therefore, the clusters should be interpreted as broad customer profiles rather than completely distinct customer types.

3. Customer Segment Profiles

The six identified segments were interpreted using their average demographic, financial and behavioral characteristics.

Cluster	Customer Segment	Customers	Avg. Age	Avg. Balance	Avg. Products	Active Member	Churn Rate
0	Active High-Balance Customers	2,505	35.45	122,091	1.31	100%	12.61%
1	Inactive Low-Balance Customers	1,728	36.34	924	1.77	0%	16.44%
2	Senior At-Risk Customers	961	60.90	77,297	1.39	88%	32.05%
3	Inactive High-Balance Customers	1,979	38.03	121,460	1.00	0%	32.44%
4	Active Low-Balance Loyal Customers	1,725	35.78	5,179	1.84	100%	8.17%
5	Multi-Product At-Risk Customers	1,102	38.21	121,451	2.24	7%	31.40%

3.1 Active High-Balance Customers

This segment contains **2,505 customers** with relatively high balances and full active membership. Their churn rate is **12.61%**, which is below the overall churn rate of 20.37%. Their financial value makes them an important group for relationship-building and loyalty initiatives.

3.2 Inactive Low-Balance Customers

This group consists of **1,728 customers** with very low average balances and inactive membership. Their churn rate is **16.44%**. Although their risk is moderate compared with the highest-risk groups, increasing engagement could help improve their relationship with the bank.

3.3 Senior At-Risk Customers

The **Senior At-Risk Customers** segment contains 961 customers with an average age of approximately **61 years**. Its churn rate is **32.05%**, making it one of the highest-risk segments. Personalized support and simplified customer services may be particularly relevant for this group.

3.4 Inactive High-Balance Customers

This is one of the most strategically important segments. It contains **1,979 customers** with an average balance of approximately **121,460** but inactive membership. The segment records the **highest churn rate of 32.44%** and accounts for **31.52% of total observed churn**. This combination of high balances, inactivity and elevated churn makes the segment a major retention priority.

3.5 Active Low-Balance Loyal Customers

This segment contains **1,725 customers** who are active but maintain relatively low balances. It has the **lowest churn rate at 8.17%**. The segment appears comparatively stable, suggesting that maintaining engagement and loyalty may be more appropriate than intensive retention interventions.

3.6 Multi-Product At-Risk Customers

The Multi-Product At-Risk segment includes **1,102 customers** with an average of approximately **2.24 products** and very low active-membership levels. Its churn rate is **31.40%**. Because these customers use multiple products but show limited activity, their relationship with the bank may require closer review and personalized engagement.

3.7 Segmentation Insight

The clustering analysis shows that churn risk is not evenly distributed across the customer base. The highest-risk profiles are primarily associated with **inactivity, older age, and certain combinations of product usage and financial characteristics**. In particular, the Inactive High-Balance segment represents the largest concentration of observed churn.

Segment Interpretation

The customer segmentation analysis divided the 10,000 customers into six distinct groups based on their demographic, financial and behavioral characteristics. The purpose of this segmentation was to understand how customer profiles differ in terms of observed churn and to identify groups that may require different retention approaches. The analysis indicates that churn is not evenly distributed across the customer base, making segment-level analysis more useful than relying only on the overall churn rate.

1. Active High-Balance Customers

This segment contains **2,505 customers** with an average balance of approximately **121,091**. All customers in this group are classified as active members, and the segment has an observed churn rate of **12.61%**, which is below the overall churn rate of 20.37%.

The relatively lower churn rate suggests that active engagement may be associated with greater customer stability. However, because customers in this segment maintain comparatively high balances, they remain financially important to the bank. Retention efforts should therefore focus on strengthening existing relationships through personalized services and loyalty-oriented offers.

2. Inactive Low-Balance Customers

The **Inactive Low-Balance Customers** segment includes **1,728 customers**. Their average balance is only around **924**, and the customers are classified as inactive. The observed churn rate is **16.44%**.

Although this rate is not among the highest in the analysis, the combination of low balances and inactivity indicates relatively limited engagement with the bank. Rather than using costly retention incentives, this group could be approached through digital engagement campaigns, relevant product information and simple opportunities to increase interaction with banking services.

3. Senior At-Risk Customers

This segment contains **961 customers** and has the highest average age among all six groups, at approximately **61 years**. Its observed churn rate is **32.05%**, placing it among the highest-risk segments.

The result is consistent with the broader age analysis, where older customer groups recorded substantially higher observed churn. This segment may benefit from personalized relationship management, accessible customer support and services designed around ease of use. The findings indicate an association between this customer profile and higher churn, rather than establishing age as a direct cause of churn.

4. Inactive High-Balance Customers

The **Inactive High-Balance Customers** segment represents **1,979 customers** and is the most important segment from a retention perspective. Customers in this group have an average balance of approximately **121,460**, but all are classified as inactive.

The segment records the **highest observed churn rate of 32.44%**. More importantly, it contributes approximately **31.52% of the total observed churn** in the dataset. This combination of high financial value, inactivity and elevated churn makes the segment a major priority for retention planning.

A suitable strategy would involve proactive re-engagement, personalized communication and targeted financial offers. The objective should be to understand the reasons for reduced activity and strengthen the existing customer relationship.

5. Active Low-Balance Loyal Customers

This group consists of **1,725 customers** with an average balance of approximately **5,179**. Despite their relatively low balances, customers in this segment are active and have the **lowest observed churn rate of 8.17%**.

The low churn rate indicates comparatively strong customer stability within this group. Therefore, aggressive retention campaigns may not be necessary. Instead, the bank can focus on maintaining engagement, improving loyalty and identifying appropriate opportunities for relationship development or cross-selling.

6. Multi-Product At-Risk Customers

The **Multi-Product At-Risk Customers** segment contains **1,102 customers** and has the highest average number of products at approximately **2.24**. However, only around **7%** of customers in this segment are active. The observed churn rate is **31.40%**.

This finding is particularly interesting because holding multiple products does not automatically correspond to lower churn in this segment. The low activity level suggests that product ownership should be considered

together with actual customer engagement. Retention teams should therefore review product usage, identify possible dissatisfaction or disengagement, and use personalized interventions where appropriate.

7. Overall Segment Interpretation

The segmentation results show that **customer engagement is an important dimension for distinguishing churn risk**. Several of the higher-risk groups are characterized by inactive membership, while the active segments generally show lower observed churn.

The most immediate priority is the **Inactive High-Balance Customers** segment because it combines a high churn rate with a large contribution to total churn. The **Senior At-Risk** and **Multi-Product At-Risk** segments also require attention because their observed churn rates exceed 31%.

Segment-Wise Churn Risk

The segment-wise churn analysis was conducted to compare the observed churn rates across the six customer segments. This analysis helps identify which customer groups require greater attention from the bank and allows retention efforts to be prioritized according to the level of observed churn.

The segments were ranked from highest to lowest churn risk as follows:

Rank	Customer Segment	Churn Rate	Risk Level
1	Inactive High-Balance Customers	32.44%	Very High
2	Senior At-Risk Customers	32.05%	Very High
3	Multi-Product At-Risk Customers	31.40%	Very High

Rank	Customer Segment	Churn Rate	Risk Level
4	Inactive Low-Balance Customers	16.44%	Medium
5	Active High-Balance Customers	12.61%	Low
6	Active Low-Balance Loyal Customers	8.17%	Low

1 Inactive High-Balance Customers

The **Inactive High-Balance Customers** segment has the highest observed churn rate at **32.44%**. These customers have relatively high balances but are classified as inactive. The combination of financial value and inactivity makes this segment particularly important from a retention perspective.

2. Senior At-Risk Customers

The **Senior At-Risk Customers** segment has a churn rate of **32.05%**, only slightly below the highest-risk segment. The segment is characterized by a relatively high average age. This indicates that customers within this profile may require more personalized support and relationship management.

3. Multi-Product At-Risk Customers

With a churn rate of **31.40%**, the Multi-Product At-Risk segment is the third-highest risk group. These customers hold multiple products but have low levels of active membership. This combination makes customer engagement an important consideration when designing retention initiatives.

4. Inactive Low-Balance Customers

The **Inactive Low-Balance Customers** segment records a churn rate of **16.44%**, placing it in the medium-risk category. Although the observed churn is considerably lower than the top three segments, the inactivity of these customers suggests that engagement-focused interventions may still be useful.

5. Active High-Balance Customers

The **Active High-Balance Customers** segment has an observed churn rate of **12.61%**. Despite having relatively high balances, these customers show lower churn than the overall customer base. Maintaining their engagement and strengthening customer relationships should therefore be prioritised over intensive recovery measures.

6. Active Low-Balance Loyal Customers

The **Active Low-Balance Loyal Customers** segment has the lowest observed churn rate at **8.17%**. Their active membership is accompanied by comparatively strong customer stability. Retention efforts for this group should focus primarily on maintaining satisfaction and engagement rather than applying aggressive retention incentives.

7. Overall Churn Risk Interpretation

The results show a clear difference in observed churn across the six customer segments. The three highest-risk groups—**Inactive High-Balance, Senior At-Risk and Multi-Product At-Risk Customers**—all have churn rates above 31%, while the lowest-risk group has a churn rate of only 8.17%.

Contribution to Total Churn

While churn rate indicates the level of risk within each customer segment, it is also important to examine how much each segment contributes to the bank’s **total observed churn**. This analysis helps identify which customer groups account for the largest share of churned customers and therefore deserve greater attention when allocating retention resources.

The dataset contains **2,037 churned customers**. The contribution of each segment was calculated based on the number of churned customers within that segment relative to the total number of churned customers.

Customer Segment	Customer Count	Churned Customers	Contribution to Total Churn
Inactive High-Balance Customers	1,979	642	31.52%
Multi-Product At-Risk Customers	1,102	346	16.99%
Active High-Balance Customers	2,505	316	15.51%
Senior At-Risk Customers	961	308	15.12%
Inactive Low-Balance Customers	1,728	284	13.94%
Active Low-Balance Loyal Customers	1,725	141	6.92%

1 Inactive High-Balance Customers

The **Inactive High-Balance Customers** segment makes the largest contribution to total churn, accounting for **642 churned customers or 31.52%** of all observed churn. This is the most significant segment from a retention perspective because it combines a relatively high churn rate with a substantial number of churned customers.

2. Multi-Product At-Risk Customers

The **Multi-Product At-Risk Customers** segment contributes **346 churned customers**, representing **16.99%** of total churn. Although the segment is smaller than some other groups, its high observed churn rate makes it an important target for customer engagement and retention activities.

3. Active High-Balance Customers

The **Active High-Balance Customers** segment accounts for **316 churned customers**, or **15.51%** of total churn. Its contribution is relatively high partly because the segment itself is the largest customer group, containing 2,505 customers. Despite its lower segment-level churn rate, its size means that it still represents a meaningful portion of total churn.

4. Senior At-Risk Customers

The Senior At-Risk segment contributes **308 churned customers**, equivalent to **15.12%** of total churn. Its relatively high churn rate, combined with its distinct age profile, makes this group relevant for targeted retention and personalized customer support.

5. Inactive Low-Balance Customers

The **Inactive Low-Balance Customers** segment contributes **284 churned customers**, representing **13.94%** of total churn. While its churn rate is lower than the three highest-risk segments, its sizeable customer population means that it still contributes noticeably to overall churn.

6. Active Low-Balance Loyal Customers

The **Active Low-Balance Loyal Customers** segment has the smallest contribution to total churn, with **141 churned customers**, representing only **6.92%** of total observed churn. This is consistent with its lowest segment-level churn rate of **8.17%**.

7. Overall Interpretation

The contribution analysis shows that **churn volume and churn rate provide different perspectives**. A segment can have a high churn rate but contribute fewer churned customers because of its smaller size. Conversely, a large segment with a moderate churn rate can still contribute substantially to total churn.

The **Inactive High-Balance Customers** segment stands out on both measures: it has the highest observed churn rate (**32.44%**) and the largest contribution to total churn (**31.52%**). Therefore, it represents the strongest immediate priority for retention efforts.

Key Findings & Discussion

The analysis of customer churn and segmentation provides several important insights into the factors associated with customer retention in the European banking dataset. The findings show that churn is not distributed equally across all customers. Instead, noticeable differences can be observed across geography, age, customer activity and customer segments.

1. Overall Churn Level

The dataset contains **10,000 customers**, of whom **2,037 have churned**, resulting in an overall churn rate of **20.37%**. This indicates that approximately one out of every five customers in the dataset has exited the bank.

The overall churn rate provides an important baseline for comparing different customer groups and identifying segments where churn is substantially higher or lower than the overall level.

2. Geographic Differences in Churn

Geography shows a clear difference in observed churn rates. **Germany records the highest churn rate at 32.44%**, compared with **16.15% in France** and **16.67% in Spain**.

The considerably higher rate observed in Germany suggests that geographic factors may be relevant when evaluating customer retention. This finding supports the need for market-specific analysis rather than assuming that the same retention pattern applies across all countries.

3. Age as an Important Churn Dimension

Age is one of the more noticeable dimensions in the analysis. The **46–55 age group has the highest observed churn rate at 50.57%**, followed by the **56–65 group at 48.32%**.

In comparison, younger customers show substantially lower churn rates. The 18–25 group records 7.53%, while the 26–35 group records 8.50%.

The correlation analysis also shows a positive relationship between age and churn, with a correlation coefficient of approximately **0.285**. This suggests that age is more strongly associated with churn than several other numerical variables examined in the dataset.

4. Customer Activity and Churn

Customer activity is another important finding. **Inactive customers have an observed churn rate of 26.85%**, compared with **14.27% among active members**.

This difference indicates that customer engagement is closely associated with retention in the dataset.

However, the analysis does not establish that inactivity directly causes customers to churn. It instead identifies inactivity as an important characteristic of several higher-risk customer profiles.

5. Financial Characteristics

Balance shows a moderate positive association with churn. Customers in the **200K+ balance group** have an **observed churn rate of 55.88%**, although the size of this group should be considered when interpreting the result.

By contrast, credit score does not show a strong separation between churned and retained customers. Churn rates across the credit-score groups remain relatively close, ranging from **19.69% to 23.64%**.

Estimated salary also provides limited differentiation, with the distributions for churned and retained customers being broadly similar. Similarly, credit-card ownership shows almost no difference, with churn rates of **20.81% for customers without a credit card** and **20.18% for those with one**.

6. Customer Segmentation Findings

The K-Means analysis divided customers into **six segments** using seven demographic, financial and behavioral variables. The segmentation revealed substantial differences in observed churn rates.

The three highest-risk segments were:

1. **Inactive High-Balance Customers — 32.44%**
2. **Senior At-Risk Customers — 32.05%**
3. **Multi-Product At-Risk Customers — 31.40%**

In contrast, **Active Low-Balance Loyal Customers** recorded the lowest churn rate at **8.17%**.

These results demonstrate why customer segmentation is useful for churn analysis. Customers with different profiles can show very different levels of observed churn even within the same banking customer base.

7. Contribution to Total Churn

The contribution analysis provides an additional perspective. The **Inactive High-Balance Customers** segment accounts for **642 churned customers**, representing **31.52% of total observed churn**.

The Multi-Product At-Risk segment contributes 16.99%, while Active High-Balance Customers account for 15.51%. This shows that the segment with the highest churn rate is also the largest contributor to overall churn in this dataset.

Therefore, retention strategies should consider both **churn rate and the number of churned customers contributed by each segment**.

8. Key Discussion

Overall, the findings indicate that **customer engagement, age, geography and specific combinations of financial and behavioral characteristics** are important dimensions in understanding churn patterns.

The analysis particularly highlights inactive and high-risk customer profiles as areas requiring greater attention. At the same time, variables such as credit-card ownership, tenure, estimated salary and credit score show comparatively weak relationships with churn in this dataset.

The results support a **targeted retention approach**, where resources are directed towards customer groups with higher observed churn and greater contribution to total churn. Such an approach can help banks move from broad retention campaigns towards more focused interventions based on customer characteristics.

Finally, the findings should be interpreted as **observational relationships rather than causal effects**. The analysis identifies patterns within the available dataset but does not establish that any single characteristic

directly causes a customer to leave. This distinction is important when translating analytical findings into business or policy decisions.

Retention Strategies & Recommendations

The findings from the churn analysis and customer segmentation indicate that retention strategies should be targeted according to customer characteristics rather than applying the same approach to the entire customer base. The analysis particularly highlights inactivity, older customer profiles and specific combinations of balance and product usage as areas requiring greater attention.

1. Re-engagement of Inactive High-Balance Customers

The **Inactive High-Balance Customers** segment should receive the highest retention priority because it has the highest observed churn rate of **32.44%** and contributes **31.52% of total observed churn**.

The bank can focus on:

- Personalized re-engagement campaigns.
- Direct communication through relationship managers.
- Targeted financial products and relevant offers.
- Regular follow-ups with customers showing declining activity.
- Understanding the reasons behind reduced account engagement.

Since these customers maintain relatively high balances, retaining their relationship may be particularly valuable to the bank.

2. Personalized Support for Senior At-Risk Customers

The **Senior At-Risk Customers** segment has an observed churn rate of **32.05%**. Retention efforts for this group should emphasize personalized and accessible customer service.

Recommended actions include:

- Dedicated relationship-management support.
- Simplified banking processes and communication.
- Proactive follow-up with less-engaged customers.
- Assistance with digital banking services where required.
- Personalized financial guidance and service interactions.

The objective should be to improve customer confidence and engagement rather than relying only on promotional incentives.

3. Engagement Strategy for Multi-Product At-Risk Customers

The **Multi-Product At-Risk Customers** segment has a churn rate of **31.40%** despite having an average of more than two products. This suggests that product ownership alone does not necessarily indicate strong customer engagement.

The bank should:

- Review the usage of individual products.
- Identify products that are no longer being used.
- Contact customers to understand service-related concerns.

- Provide personalized product support.
- Offer relevant solutions based on actual customer needs.

A customer having multiple products should therefore not automatically be considered a low-risk customer.

4. Improving Engagement Among Inactive Low-Balance Customers

The **Inactive Low-Balance Customers** segment has a moderate churn rate of **16.44%**. Because these customers have relatively low balances, retention initiatives should remain cost-effective.

Possible actions include:

- Digital engagement campaigns.
- Personalized notifications and relevant offers.
- Financial product education.
- Encouraging regular use of banking services.
- Low-cost loyalty and engagement programs.

The focus should be on increasing engagement without allocating excessive retention resources to this comparatively lower-risk segment.

5. Maintaining Relationships with Active High-Balance Customers

The **Active High-Balance Customers** segment records a relatively lower churn rate of **12.61%**, but it contains a large number of customers and represents **15.51% of total observed churn**.

The bank should therefore focus on maintaining their existing relationship through:

- Premium customer services.

- Personalized financial offers.
- Loyalty programs.
- Regular relationship-manager interactions.
- Relevant cross-selling opportunities.

The objective should be preventive retention rather than aggressive intervention.

6. Maintaining Loyalty Among Active Low-Balance Customers

The **Active Low-Balance Loyal Customers** segment has the lowest observed churn rate at **8.17%**. Since this group already demonstrates comparatively strong retention, extensive intervention may not be necessary.

The bank can focus on:

- Maintaining service quality.
- Continuing customer engagement.
- Rewarding loyalty.
- Providing relevant product recommendations.
- Identifying suitable opportunities for relationship development.

This allows the bank to preserve customer loyalty while avoiding unnecessary retention expenditure.

7. Priority-Based Retention Framework

Based on the analysis, retention resources can be prioritized as follows:

Priority	Customer Segment	Main Strategy
Very High	Inactive High-Balance Customers	Re-engagement and personalized retention
Very High	Senior At-Risk Customers	Personalized support and relationship management
Very High	Multi-Product At-Risk Customers	Product-usage review and targeted engagement
Medium	Inactive Low-Balance Customers	Cost-effective digital engagement
Low	Active High-Balance Customers	Loyalty and relationship strengthening
Low	Active Low-Balance Loyal Customers	Maintain engagement and loyalty

8. Overall Recommendations

The analysis suggests that the bank should adopt a **segmentation-based retention strategy**. The highest priority should be given to customers who combine high observed churn with characteristics that indicate reduced engagement or higher customer risk.

The bank should particularly focus on **Inactive High-Balance Customers**, followed by **Senior At-Risk** and **Multi-Product At-Risk Customers**. At the same time, lower-risk segments should be managed through relationship maintenance and loyalty-building rather than costly retention campaigns.

A practical retention program should also continuously monitor customer activity and segment movement. Customers whose behavior changes over time can then be reassessed and moved into more appropriate retention categories.

It is important to note that these recommendations are based on **patterns and associations identified in the available dataset**. They should therefore be treated as targeted business recommendations for further evaluation rather than evidence that a particular customer characteristic directly causes churn.

Limitations

Although the analysis provides useful insights into customer churn and segmentation, several limitations should be considered when interpreting the results.

1. Dataset Scope

The analysis is based on a dataset of **10,000 European banking customers**. Since the analysis is limited to the available dataset, the findings may not represent the behavior of all banking customers across Europe or other banking markets.

2. Limited Time Dimension

The dataset contains customer information for a specific period, while the Year variable is constant at **2025**.

Therefore, the analysis does not capture how customer behavior or churn patterns change over multiple years.

3. Limited Customer Variables

The dataset contains demographic, financial and behavioral variables such as age, balance, credit score, tenure, products and activity status. However, other potentially relevant factors, such as customer satisfaction, service quality, transaction frequency, complaints, interest rates, economic conditions and reasons for leaving, are not available.

As a result, the analysis cannot provide a complete explanation of why customers churn.

4. Limitations of K-Means Segmentation

Customer segmentation was performed using the **K-Means clustering algorithm**. The analysis selected six clusters based on the highest silhouette score of **0.165**. However, the relatively low silhouette score indicates that the clusters are not strongly separated.

Therefore, the six segments should be viewed as useful analytical profiles rather than completely distinct customer groups.

5. Small Group Sizes in Some EDA Categories

Some categories showed unusually high churn rates, particularly customers with **three or four products** and those with balances above **200K**. These categories may contain relatively fewer observations compared with the overall dataset.

Therefore, their high churn rates should be interpreted carefully and should not be generalized without checking the underlying sample sizes.

6. Correlation Does Not Imply Causation

The correlation analysis identifies relationships between variables and churn, but correlation does not establish cause and effect. For example, the positive relationship between age and churn does not mean that age itself directly causes customers to leave.

Similarly, the association between inactivity and higher churn should not automatically be interpreted as proof that inactivity causes churn.

7. Model and Analysis Limitations

The project primarily focuses on **exploratory analysis and K-Means segmentation** rather than developing and validating a predictive churn model. Consequently, the analysis identifies customer patterns and risk groups but does not provide an individual-level probability of future churn.

8. Generalizability of Recommendations

The retention recommendations are based on patterns identified within the available dataset. Their effectiveness would need to be evaluated using actual customer behavior and business outcomes before being implemented on a large scale.

Future Scope

The present study provides a useful foundation for understanding customer churn through exploratory analysis and customer segmentation. However, the analysis can be extended in several ways to make the findings more detailed and useful for practical banking decision-making.

1. Development of Predictive Churn Models

Future studies can use machine learning techniques such as **Logistic Regression, Random Forest, Decision Trees and other classification models** to predict the probability of individual customers churning. This would allow banks to identify customers who may be at higher risk before they actually leave.

2. Use of Longitudinal Customer Data

The current dataset represents a limited time period. Future research could use **multi-year customer data** to examine how customer behavior changes over time. Tracking changes in account activity, balance, product usage and engagement could provide a better understanding of the development of churn risk.

3. Incorporating Additional Customer Behavior Variables

Future analysis could include variables such as **transaction frequency, digital banking usage, customer complaints, service interactions, loan behavior and customer satisfaction**. These variables may provide additional information that is not captured by the current dataset.

4. Advanced Customer Segmentation

The segmentation can be further explored using alternative clustering techniques such as **Hierarchical Clustering, DBSCAN or Gaussian Mixture Models**. Comparing different approaches may help determine whether more clearly separated customer profiles can be identified.

5. Real-Time Churn Monitoring

The existing Stream lit dashboard provides interactive analysis of customer churn and segments. In the future, it could be enhanced to support **real-time or regularly updated customer data**, allowing banking teams to monitor changes in churn patterns and segment-level risk more continuously.

6. Personalized Retention Strategies

Future work can move from segment-level recommendations towards **individual-level retention strategies**. Customers could be assigned personalized interventions based on their predicted churn probability, engagement level, financial profile and product usage.

7. Evaluation of Retention Strategies

An important future step would be to measure whether the recommended retention strategies actually reduce churn. Banks could evaluate interventions using controlled experiments or before-and-after comparisons to determine which strategies provide the greatest improvement in customer retention.

8. Geographic Expansion

Since the current analysis focuses on three European markets, future studies could include **additional countries and regions**. This would help determine whether the observed churn patterns are specific to the current markets or represent broader European banking trends.

9. Integration of External Factors

Future research could also consider external factors such as **economic conditions, interest rates, inflation and changes in the banking environment**. Incorporating these factors may provide a broader explanation of changes in customer churn over time.

Conclusion

This study examined customer churn patterns and customer segments in the European banking dataset using exploratory data analysis and K-Means clustering. The analysis was conducted on **10,000 customers**, of whom **2,037 had churned**, resulting in an overall churn rate of **20.37%**.

The exploratory analysis showed noticeable differences in churn across customer characteristics. **Germany recorded the highest geographic churn rate at 32.44%**, while customers aged **46–55 recorded a churn rate of 50.57%**. Customer activity also showed an important difference, with inactive customers having a churn rate of **26.85%**, compared with **14.27% for active customers**. These patterns indicate that customer engagement, age and geography are useful dimensions for understanding differences in observed churn.

Customer segmentation further divided the customer base into **six profiles** using seven demographic, financial and behavioral variables. Among these segments, **Inactive High-Balance Customers** had the highest observed churn rate at **32.44%** and contributed **31.52% of total observed churn**. **Senior At-Risk Customers** and **Multi-Product At-Risk Customers** also showed high churn rates of **32.05%** and **31.40%**, respectively. In contrast, **Active Low-Balance Loyal Customers** had the lowest observed churn rate at **8.17%**.

The findings suggest that banks can benefit from moving towards a **segment-specific retention approach**.

Greater attention should be given to inactive and high-risk customer groups, while lower-risk segments can be managed through continued engagement and loyalty-building activities. The Stream lit dashboard developed as part of the project provides an interactive way to examine these customer segments and churn patterns.

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You can use the following references in your research paper. I've kept them relevant to the methods and tools actually used in your project.

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